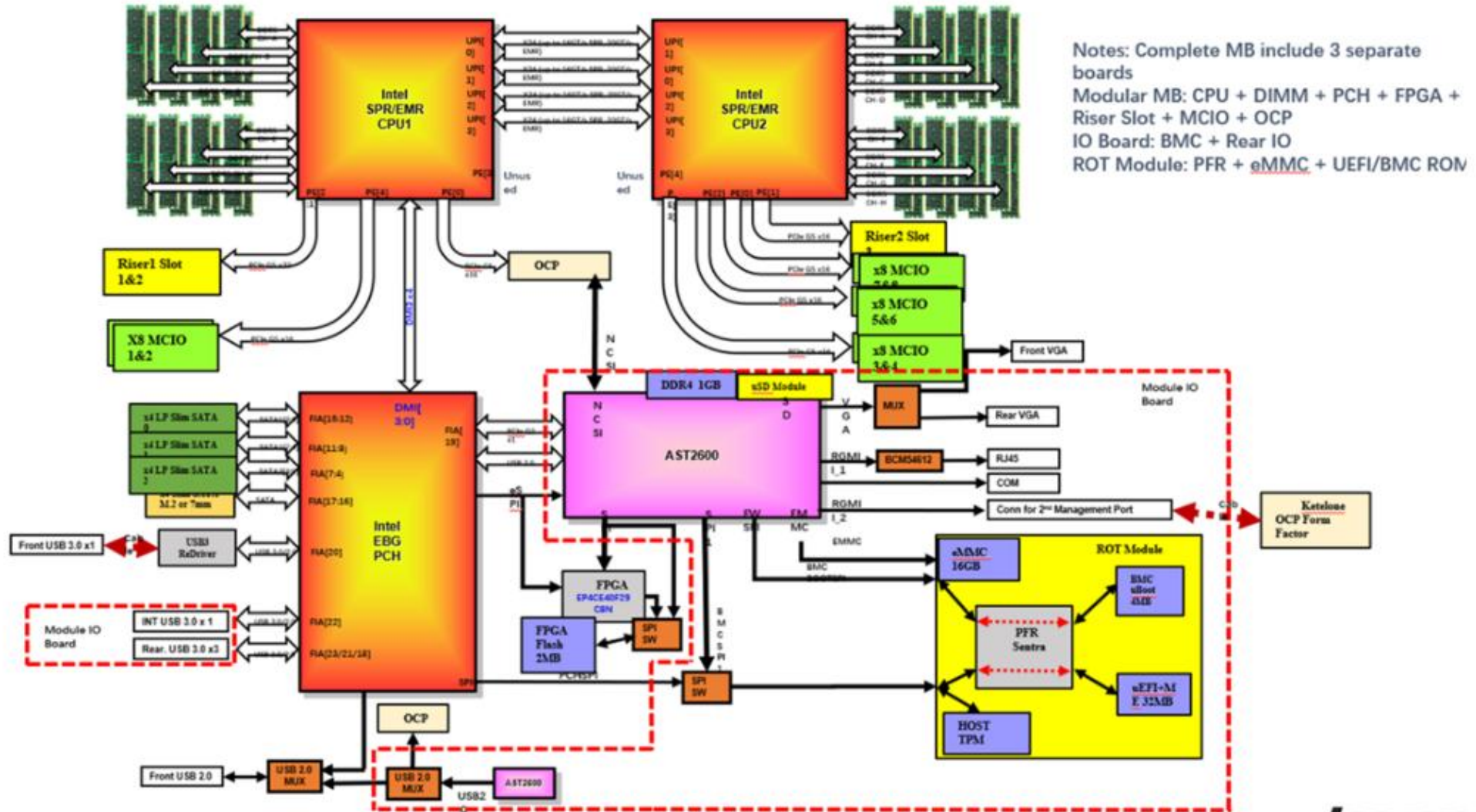


System configurations and diagrams

System block diagram and hardware configurations

Lenovo

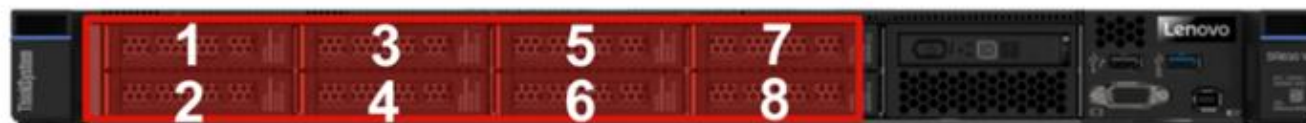
SR630 V3 system block diagram



Front drive bay configurations -1

The SR630 V3 front drive bay supports the following configurations:

- 3.5-inch hot-swap drive bays
 - Four SAS/SATA
 - Four AnyBay (PCIe 4.0)
- Four 2.5-inch hot-swap drive bays
 - Four SAS/SATA
 - Four NVMe (PCIe 4.0)
- Eight 2.5-inch hot-swap drive bays
 - Eight SAS/SATA



Front drive bay configurations -2

The SR630 V3 front drive bay supports the following configurations:

- 10 2.5-inch hot-swap drive bays
 - 10 SAS/SATA
 - Six SAS/SATA plus four AnyBay (PCIe 4.0 or 5.0)
 - Six SAS/SATA plus four NVMe (PCIe 4.0)
 - Six SAS/SATA, two AnyBay, plus two NVMe (PCIe 4.0 or 5.0)
 - 10 AnyBay (PCIe 4.0 or 5.0)
 - 10 NVMe (PCIe 4.0 or 5.0)
- EDSFF drive bays
 - 16 E1.S (9.5 mm or 15 mm) hot-swap NVMe drive bays (PCIe 4.0)



Rear drive bay configurations

The SR630 V3 rear drive bay supports the following configurations:

- Two 2.5-inch hot-swap SAS/SATA drive bays
- Two 2.5-inch hot-swap NVMe drive bays (PCIe 4.0)
- Two 7 mm SAS/SATA drive bays
- Two 7 mm NVMe drive bays (PCIe 4.0)



M.2 and 7 mm drive boot solution configurations

The SR630 V3 supports one or two M.2 or 7 mm SATA or NVMe drives for use as an operating system boot solution or as additional storage.

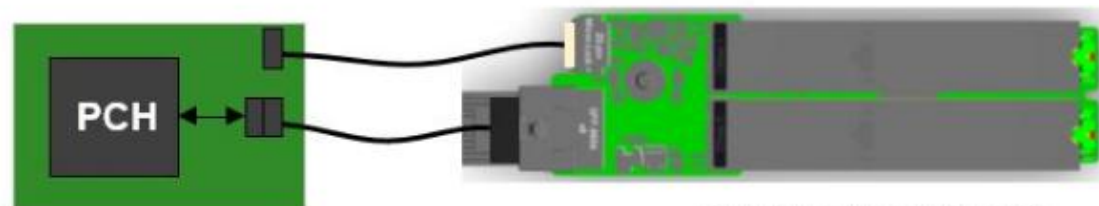
Configuration notes:

- RAID support is implemented using VROC (no adapter needed) or with the use of an additional RAID adapter.
- If RAID is enabled using VROC, use the following features:
 - VROC SATA support: Onboard SATA software RAID mode for 7 mm drives
 - VROC NVMe support: Intel VROC (VMD NVMe RAID) standard for 7 mm drives
- If RAID is enabled using a RAID adapter, the adapter must be installed in PCIe slot 1 or 2
 - RAID support for 7 mm SATA drives requires a RAID 5350-8i adapter
 - RAID support for 7 mm NVMe drives requires a RAID 540-8i adapter operating in Tri-Mode
- RAID can be enabled using an M.2 or 7 mm adapter through an onboard Marvell controller.
- M.2 and 7 mm drives are not supported together in the same configuration.

M.2 boot solution configurations

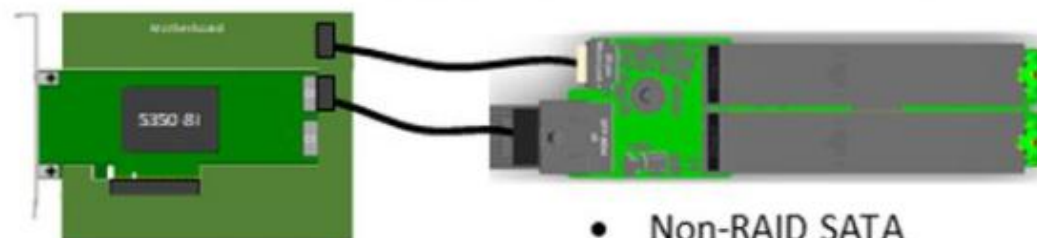
The following figures show M.2 drives connected either to the system board or to a RAID adapter. Note that different configurations use different cables. Refer to [Lenovo Docs](#) for more information.

- SATA Boot



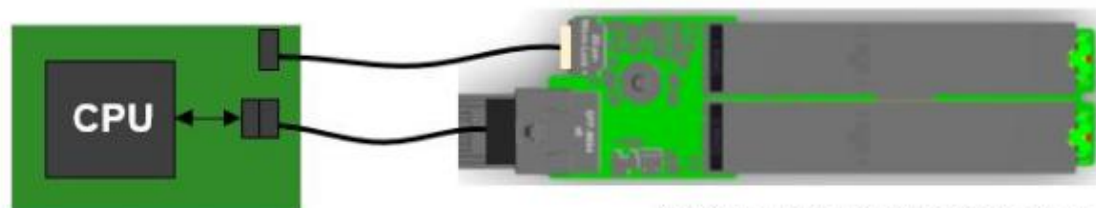
- SATA RAID (VROC)
- Non-RAID SATA

- Hardware RAID (SFF/CFF 5350-8i SATA RAID)



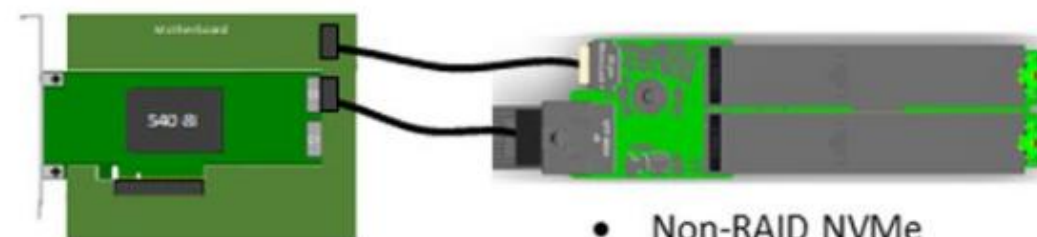
- Non-RAID SATA

- NVMe Boot (CPU attached)



- NVMe RAID (VROC standard)
- Non-RAID NVMe

- Hardware RAID (SFF 540-8i NVMe RAID)



- Non-RAID NVMe
- NVMe RAID

Note: For the latest lists of supported M.2 drives and adapters, refer to the [Lenovo Press product guide](#).

7 mm boot solution configurations

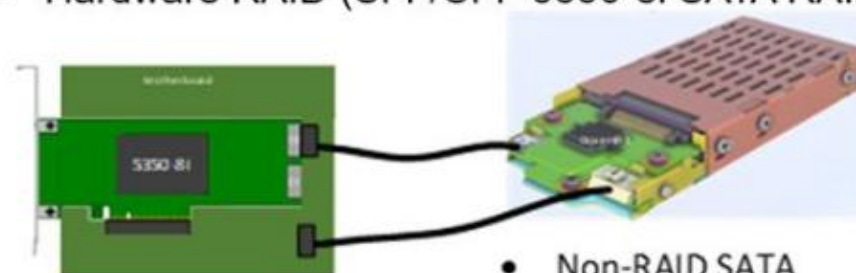
The following figures show 7 mm drives connected either to the system board or to a RAID adapter. Note that different configurations use different cables. Refer to [Lenovo Docs](#) for more information.

- SATA Boot



- SATA RAID (VROC)
- Non-RAID SATA

- Hardware RAID (SFF/CFF 5350-8i SATA RAID)



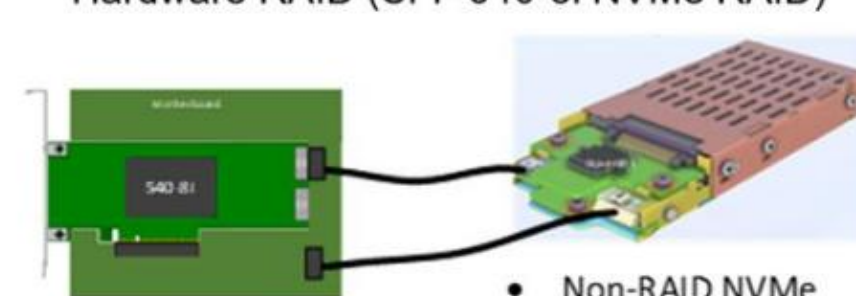
- Non-RAID SATA
- SATA RAID

- NVMe Boot (CPU attached)



- NVMe RAID (VROC standard)

- Hardware RAID (SFF 540-8i NVMe RAID)



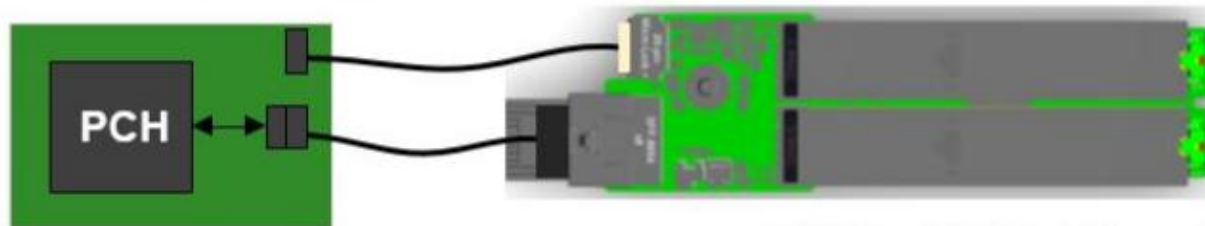
- Non-RAID NVMe
- NVMe RAID

Note: For the latest list of supported 7 mm drives and RAID adapters, refer to the [Lenovo Press product guide](#).

M.2 or 7 mm boot solution configurations with a Marvell controller

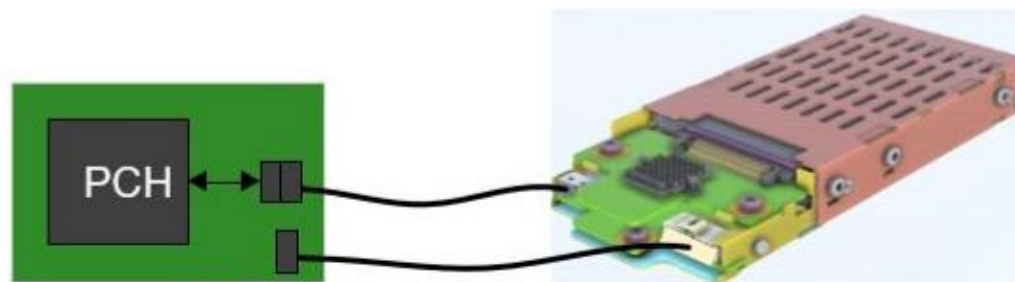
The following figures show M.2 or 7 mm drives with a Marvell controller connected to the system board for hardware RAID. Note that different configurations use different cables. Refer to [Lenovo Docs](#) for more information.

- M.2 NVMe Boot



- NVMe RAID (Marvell controller)

- 7 mm NVMe Boot



- NVMe RAID (Marvell controller)

I/O expansion slot configurations -1

The SR630 V3 supports a total of up to 3 PCIe slots, all with rear access, plus a dedicated OCP 3.0 SFF slot for networking.

A



- Three rear PCIe slots
- All x16 low profile (LP)

B



- Two rear PCIe slots
- Slot 1: x16 LP
 - Slot 2: x16 FHHL

C



- Two rear PCIe slots
- Two x16 LP

I/O expansion slot configurations -2

The SR630 V3 supports a total of up to 3 PCIe slots, all with rear access, plus a dedicated OCP 3.0 SFF slot for networking.



One rear PCIe slot

- x16 LP



Two rear PCIe slots

- Slot 1: x16 LP
- Slot 2: x16 FHHL



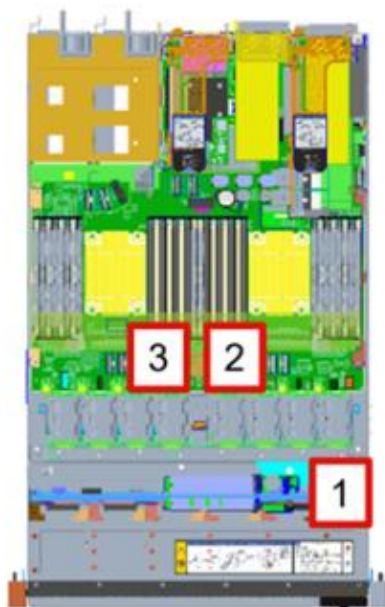
Two front PCIe slots

- Slot 4: x16 LP
- Slot 5: x16 FH

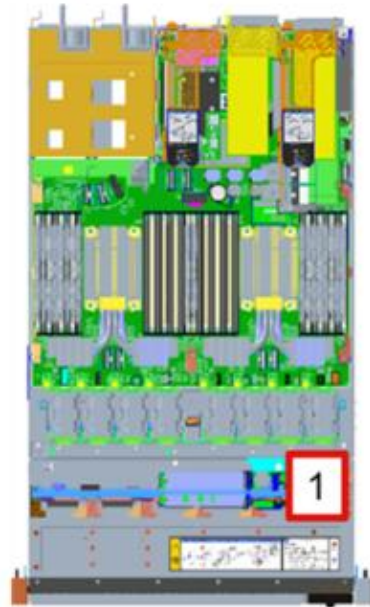
Note: The front PCIe adapter and OCP configuration is scheduled to be available in Q2 2023.

RAID flash power module location

The location of the RAID flash power modules (supercaps) depends on the configuration used in the server. The modules can be located on the air baffle, the front of the chassis, or in a PCIe slot.



2.5-inch model with
standard heat sink



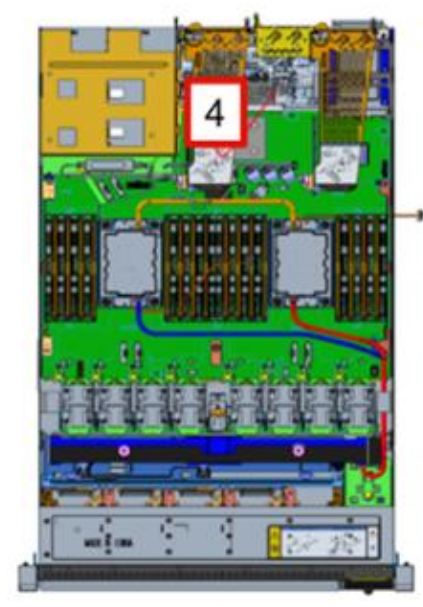
2.5-inch model with
performance heat
sink



3.5-inch model with
standard heat sink



3.5-inch model with
performance heat
sink



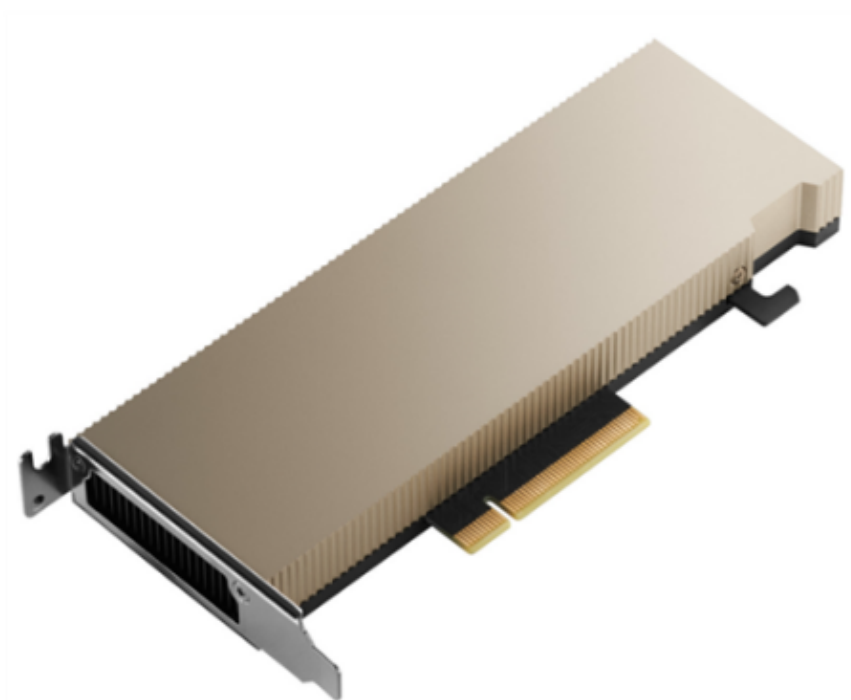
Closed loop model

GPU adapters

The SR630 V3 supports the following GPU in PCIe slots:

- ThinkSystem NVIDIA A2 16GB PCIe Gen4 Passive GPU

Note: For the NVIDIA A2, or any other passively cooled GPU (a GPU without an integrated fan), performance fans are required. For more information about these GPUs, refer to the [ThinkSystem GPU Summary](#).



Memory options

The SR630 V3 supports 16 DIMMs per processor. Each processor has eight memory channels with two DIMMs per channel (2DPC). There is support for 1DPC at 4800 MHZ and 2DPC at 4400 MHZ.

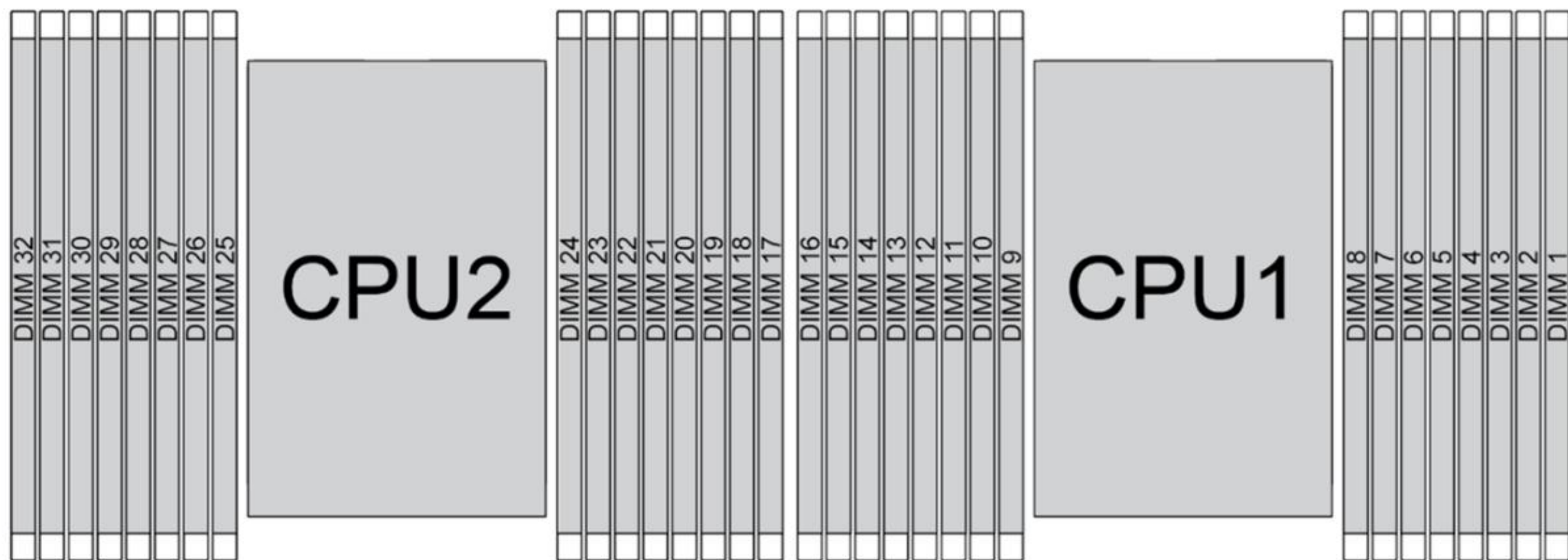
The SR630 V3 supports two memory modes:

- Independent memory mode
- Memory mirroring mode

Note: For detailed memory configuration and installation rules, refer to the [Memory module installation rules and order](#) section of the SR630 V3 Setup Guide.

Memory slot and channel identification

The following figure shows the layout of the memory modules and processors. Click [HERE](#) to see the memory slot and channel identification.



Memory slot and channel identification



The following figure shows the layout of the memory modules and processors. Click [HERE](#) to see the memory slot and channel identification.

Processor	CPU 1															
Controller	iMC3				iMC2				iMC0				iMC1			
Channel	CH1		CH0		CH1		CH0		CH0		CH1		CH0		CH1	
Slot No.	0	1	0	1	0	1	0	1	1	0	1	0	1	0	1	0
DIMM No.	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
Processor	CPU 2															
Controller	iMC3				iMC2				iMC0				iMC1			
Channel	CH1		CH0		CH1		CH0		CH0		CH1		CH0		CH1	
Slot No.	0	1	0	1	0	1	0	1	1	0	1	0	1	0	1	0
DIMM No.	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17